

Frequency and Risk Factors of Kidney Alterations in Children and Adolescents who Are Overweight and Obese in a Primary Health-care Setting

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Objective: Obesity is clearly associated to kidney disease in adult population; however, there is scarce evidence in children and adolescents. The aim was to compare frequency of renal damage according to the presence of overweight-obesity in children and adolescents, as well as to compare nutritional and biochemical risk factors, according to the presence of kidney alterations.

Methods: Cross-sectional study; 172 children and adolescents, 6-16 years old, without malnutrition, diabetes mellitus, hypertension and independent comorbid conditions associated to obesity or kidney disease, as well as transitory causes of microalbuminuria (MA) from a Primary Health-Care Unit were included. Clinical, biochemical, anthropometric and dietetic evaluations were measured in all subjects; subsequently they were classified as normal weight, overweight and obesity groups according to sex- and age-adjusted body mass index (BMI). Glomerular filtration rate (GFR, estimated by Schwartz equation) and albuminuria (albumin/creatinine ratio) were determined. Presence of kidney alterations was measured as decreased GFR (<90 mL/min/1.73m²), hyperfiltration (>170 mL/min/1.73m²) and MA (30-300 mg/g).

Results: Compared with controls, subjects with overweight-obesity had significantly ($P<.05$) abdominal obesity (0 vs 69%), hypertension (19 vs 26%), hypertriglyceridemia (11 vs 47%), high low-density lipoprotein cholesterol (2 vs 8%) and low high-density lipoprotein cholesterol (HDL-cholesterol; 2 vs 28%), hyperuricemia (11 vs 28%) and hyperinsulinemia (8 vs 70%). Hyperfiltration and MA were present in 5 and 4 subjects with overweight/obesity, respectively, whereas decreased GFR was present in only 1 subject with obesity. Normal weight subjects had no kidney alterations. In multivariate analysis, kidney alterations were significantly predicted by higher BMI and lower HDL-cholesterol.

Conclusions: Kidney alterations were observed only in subjects with overweight (3.6%) and obesity (9.9%), who additionally, displayed cardiometabolic and kidney disease risk factors more frequently than normal weight subjects.

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Introduction

OBESITY IS A worldwide public health problem and is progressively increasing; in Mexico, the prevalence of overweight and obesity is extremely high in both adult (72%) and pediatric (35%) population.¹

In adults, obesity and its related complications (hyperglycemia, hypertension, and dyslipidemia) are strongly associated to kidney disease, evidenced by microalbuminuria (MA), high serum creatinine, and decreased glomerular filtration rate (GFR).²⁻⁴ Obesity is also associated to glomerular hyperfiltration and increased renal plasma flow.^{4,5} Moreover, central obesity seems to be more important than obesity itself when predicting kidney damage, as patients with abdominal obesity had greater albuminuria and lower creatinine clearance than those with peripheral fat distribution.⁶ On the other hand, beneficial effects on kidney function have been observed after weight reduction in subjects with obesity, evidenced by improvement in glomerular hyperfiltration and decrease in renal plasma flow.⁷

Physiopathologic mechanisms of obesity-linked kidney disease have been described^{8,9}: obesity increases blood pressure (BP), renal vasodilatation, and glomerular filtration neurohumoral activation, as well as metabolic and

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inflammatory changes may cause glomerular and tubular damages and elevation of pressure natriuresis, which results in enhancement of hypertension and hyperfiltration, preceding the gradual loss of kidney function. The added physical compression of visceral and perinephritic fat deposition may exacerbate fibrosis and microvascular damages in the kidney.

In spite of the aforementioned information, the association of obesity with kidney alterations in childhood and adolescence has not been completely elucidated, and there are no data in Latin-American population (where the problem of obesity is remarkable).¹⁰

Therefore, the aim of this study was to compare the frequency of kidney damage according to the presence of overweight/obesity in children and adolescents, as well as to compare dietetic, anthropometric, and biochemical risk factors according to the presence of kidney alterations.

Methods

This cross-sectional study included a total of 172 children and adolescents aged 6–18 years attending the PREVENIMSS program in a primary health-care unit. PREVENIMSS is a program managed by the Mexican Institute of Social Security to promote the prevention of chronic noncommunicable diseases. Subjects with malnutrition, diabetes mellitus, hypertension, and independent comorbid conditions associated to obesity or kidney disease, as well as those with transitory causes of MA, were excluded. Individuals were also excluded if they had hematuria or infection evidence in the first voiding urine sample. Participants were finally included after signing a form of parental informed consent.

Subjects were classified according to body mass index (BMI) into 3 groups: (1) normal weight, (2) overweight, and (3) obesity; as BMI varies with age and gender, we used the International Obesity Task Force criteria that recommend cutoffs based on age- and gender-specific values that project to the adult cutoffs of 25 kg/m² for overweight and 30 kg/m² for obesity as recommended for epidemiological studies.¹¹ All subjects had clinical, biochemical, anthropometric, and dietetic evaluations.

BP was measured in 3 separated occasions using an appropriate cuff for the arm size of children and was classified as normal (BP < 90th percentile for children aged between 1 and 13 years and <120/<80 mmHg for children aged ≥13 years), elevated BP (≥90th percentile to <95th percentile for children aged between 1 and 13 years and 120/<80 mmHg to 129/<80 mmHg for children aged ≥13 years), or hypertension (≥95th percentile for children aged between 1 and 13 years and ≥130/80 mmHg for children aged ≥13 years), adjusted for gender, age, and height, according to current recommendations.¹² A fasting blood sample was obtained to determine hemoglobin; glucose; high-density lipoprotein cholesterol (HDL), low-density lipoprotein cholesterol

(LDL), and total cholesterol; triglycerides; uric acid; and creatinine by habitual techniques. Serum insulin was measured by radioimmunoassay in the same sample. Serum creatinine was used for GFR estimation with the Schwartz formula, as recommended for pediatric population.¹³ GFR was classified as normal (90–169 mL/min/1.73 m²), slightly decreased (60–89 mL/min/1.73 m²), and hyperfiltration (≥170 mL/min/1.73 m²).¹³ In a second urine sample, albuminuria was determined by means of nephelometry (Behring Nephelometer Analyzer II; Behring Diagnostics GmbH, Marburg, Germany) and adjusted to creatinuria. MA was defined as an albumin-creatinine ratio of 30–300 mg/g and macroalbuminuria as >300 mg/g.¹³ Kidney alterations were defined as the presence of decreased GFR, hyperfiltration, or MA.

Anthropometric evaluation consisted of measurement of triceps and subscapular skinfolds and mid-arm and waist circumference. A waist circumference ≥90th percentile, adjusted for gender, age, and race, was considered abdominal obesity.¹⁴ Body composition (body fat percentage, lean body mass, and total corporal water) was estimated with bioelectrical impedance (Tanita TBF-410; Tokyo, Japan). Nutrient intake was calculated from a 24-hour dietary recall and analyzed in a computerized database containing the details of nutritional values of the main consumed Mexican foods (*Nutripac*® 1.5, Mexico, DF; www.nutripac.com.mx).

This study adheres to the Declaration of Helsinki and was approved by the local committee of research and ethics (No. 085/2005).

Statistical Analysis

Data are presented as mean ± standard deviation, median (percentiles 25–75%), or percentages as appropriate. Comparisons between obesity, overweight, and normal-weight groups were made by means of one-way analysis of variance or χ^2 test. For comparisons between groups with and without kidney alterations, Mann-Whitney U and Fisher's exact tests were used. A Spearman or Pearson correlation analysis was used as appropriate. Logistic regression analysis was used to identify variables predicting kidney alterations. A *P* value < .05 was considered significant.

Results

One-hundred seventy-two children and adolescents were included in the study; 46 (27%) were classified as normal weight, 55 (32%) overweight, and 71 (41%) obesity. Comparisons of demographic and clinical characteristics between groups are shown in Table 1. Subjects with obesity had a significantly higher proportion of family history of obesity and higher systolic BP than the others. A nonsignificant trend to higher birth weight, diastolic BP, and sedentary activities was observed in individuals with obesity. No other differences were observed regarding

Table 1. Comparison of Demographic and Clinical Characteristics According to Body Mass Index

Variable	Normal Weight, N = 46	Overweight, N = 55	Obesity, N = 71
Age (years)	10.5 ± 3.5	11.3 ± 2.9	10.4 ± 2.6
Female sex, n (%)	21 (46)	26 (47)	26 (37)
Family history, n (%)			
Obesity	25 (56)	36 (78)*	59 (83)*
Diabetes	37 (82)	33 (73)	60 (84)
Hypertension	28 (62)	33 (72)	48 (69)
Kidney disease	13 (29)	9 (20)	15 (21)
Birth weight (g)	3237 ± 527	3236 ± 526	3364 ± 600
Sedentary activities (h/day)	4.3 ± 2.0	4.3 ± 2.1	4.6 ± 2.7
Systolic BP (mmHg)	101 ± 11	104 ± 9	109 ± 11*,†
Diastolic BP (mmHg)	66 ± 10	66 ± 7	70 ± 9

BP, blood pressure.

**P* < .05 versus normal weight.†*P* < .05 versus overweight.

age, gender, and family history of diabetes, hypertension, or kidney disease.

Biochemical and kidney function characteristics between groups are shown in Table 2. Regarding biochemical variables, the higher the BMI, the lower the glucose and HDL cholesterol and the higher the insulin, uric acid, and triglycerides levels. No statistical differences were found in the case of total and LDL cholesterol. Prevalence of risk factors for kidney disease was significantly higher in overweight and obesity groups (Figure 1). In children and adolescents with overweight/obesity, individual prevalence of the risk factors such as dyslipidemia, hyperinsulinemia, and abdominal obesity was >60%, whereas hyperuricemia, hypertension, hypertriglyceridemia, and low HDL cholesterol were present in about 30% each.

Regarding kidney function and urinary albumin excretion, no significant differences were found in serum creatinine, GFR, and albuminuria between groups (Table 2). However, 1 case of decreased GFR and 4 cases of hyperfiltration were found in the obesity group, and 1 case of hyperfiltration, in the overweight group. In addition, MA was present in 4 cases (none of these cases had hypertension), 1 in the overweight group and 3 in the obesity group. Considering both GFR alterations and MA together, the frequency of kidney alterations was almost 10% in obesity, 4% in overweight, and 0% in normal-weight groups (Figure 2). The only case with decreased GFR also presented MA but not hypertension. From those subjects with kidney alterations, 3 cases with hyperfiltration (all of them with obesity) also presented hypertension but not MA.

In a further analysis, children and adolescents were classified according to the presence of kidney alterations (decreased GFR, hyperfiltration, and MA) (Tables 3 and

Table 2. Comparison of Biochemical and Kidney Function Characteristics According to Body Mass Index

Variable	Normal Weight, N = 46	Overweight, N = 55	Obesity, N = 71
Hemoglobin (g/dL)	14.3 ± 1.0	14.2 ± 0.9	14.3 ± 0.9
Glucose (mg/dL)	88 ± 9	85 ± 10	84 ± 9*
Insulin (μIU/mL)	9 ± 4	15 ± 7*	21 ± 13*,†
Uric acid (mg/dL)	4.1 ± 1.1	4.6 ± 1.4	5.1 ± 1.2*
Total cholesterol (mg/dL)	153 ± 25	164 ± 31	164 ± 33
LDL cholesterol (mg/dL)	84 ± 21	93 ± 27	95 ± 24
HDL cholesterol (mg/dL)	51 ± 13	41 ± 10*	39 ± 8*
Triglycerides (mg/dL)	92 ± 29	141 ± 70*	154 ± 90*
Creatinine (mg/dL)	0.63 ± 0.16	0.66 ± 0.16	0.63 ± 0.13
GFR (mL/min/1.73 m ²)	129 ± 21	129 ± 21	133 ± 22
Decreased GFR, n (%)	0 (0)	0 (0)	1 (1.4)
Hyperfiltration, n (%)	0 (0)	1 (1.8)	4 (5.6)
Albuminuria (mg/g)	4.7 (3.4-6.7)	5.1 (3.5-7.5)	5.0 (3.6-7.7)
Microalbuminuria, n (%)	0 (0)	1 (1.8)	3 (4.3)
Macroalbuminuria, n (%)	0 (0)	0 (0)	0 (0)

GFR, glomerular filtration rate; LDL, low-density lipoprotein; HDL, high-density lipoprotein.

Conversion factors for units: glucose in mg/dL to mmol/L, ×0.05551; uric acid in mg/dL to μmol/L, ×59.48; cholesterol in mg/dL to mmol/L, ×0.02586; triglycerides in mg/dL to mmol/L, ×0.01129; creatinine in mg/dL to μmol/L, ×88.4.

**P* < .05 versus normal weight.†*P* < .05 versus overweight.

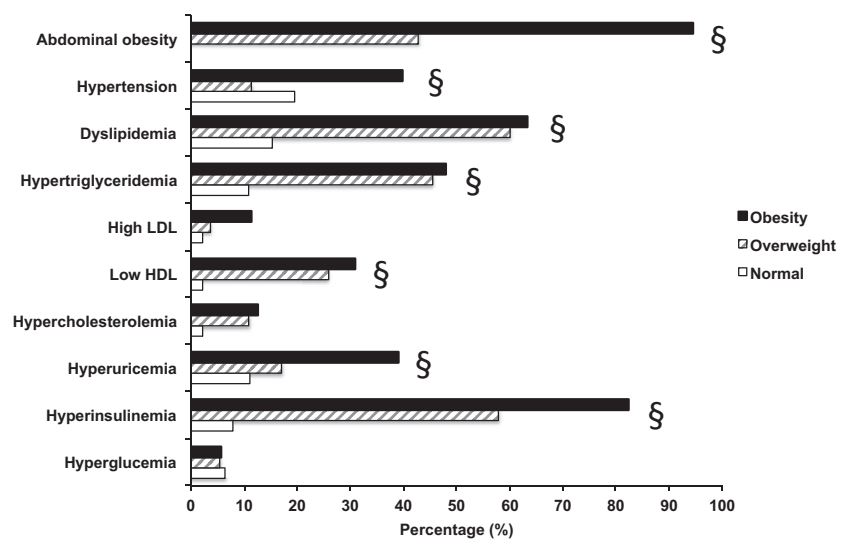
4). Those subjects with kidney alterations had a significant lower HDL cholesterol level and a nonsignificant trend to lower birth weight than subjects without kidney alterations; no differences on age, sex, blood pressure, and other biochemical variables were found between groups. Subscapular skinfold thickness, body fat percentage, and abdominal obesity were significantly higher for subjects with kidney alterations than those in the other group.

Correlation and Multivariate Analysis Results

A negative correlation was found between GFR and age (*r*: -0.37, *P* < .0001), uric acid (*r*: -0.37, *P* < .0001), and monounsaturated fatty acid intake (%) (*r*: -0.15, *P* = .06). In addition, age (*r*: -0.32, *P* < .0001) and uric acid levels (*r*: -0.21, *P* = .007) negatively correlated with albuminuria, whereas sodium intake (*r*: 0.16, *P* = .04) and protein intake (*r*: 0.19, *P* = .03) had a positive correlation.

In the logistic regression analysis (Cox & Snell *R*²: 0.18, *P* = .002), the presence of kidney alterations was significantly predicted by higher BMI (odds ratio = 10.6

Figure 1. Prevalence of risk factors for kidney disease according to body mass index. HDL, LDL. § $P < .05$ between groups. HDL, high-density lipoprotein; LDL, low-density lipoprotein.



[95% confidence interval: 1.1–103], $P = .04$) and lower HDL cholesterol (odds ratio = 9.1 [95% confidence interval: 1.0–82], $P = .05$). Other independent variables included in this analysis were age, systolic and diastolic BP, waist circumference, uric acid, LDL cholesterol, insulin, and sedentary activity.

Discussion

This study has shown that kidney alterations are more frequent in children and adolescents with overweight/obesity than in those with normal BMI.

Obesity is well recognized as to be associated to multiple metabolic disorders, both in adult and pediatric population. In our study, children and adolescents with overweight/obesity had higher systolic BP and metabolic alterations than those with normal weight, which is consistent with literature.¹⁵ In contrast with some international series (but consistent with other Mexican studies), a high proportion of hypertriglyceridemia and

low HDL cholesterol were observed in subjects with obesity, which may be explained by inadequate dietary patterns, including a high intake of sugars commonly seen in our setting.¹⁶

Kidney disease has been associated with obesity, particularly in adults²; however, scarce information exists in pediatric population. A younger age has been frequently reported in patients with end-stage renal disease in emerging countries than in developed countries.¹⁷ In populations with epidemiological nutritional transition—with a change toward an increase in morbidity and mortality due to noncommunicable diseases, coincidentally with an increase in intake of sugar- and fat-rich ultraprocessed food and sedentary lifestyles—it is possible that kidney alterations start early in life, and thus, high-risk subjects such as children and adolescents with obesity may have an undiagnosed kidney disease, which might be manifested in adulthood as kidney failure. In a large longitudinal cohort, Silverwood et al. found that

Figure 2. Frequency of kidney alterations according to body mass index ($P = .05$ between groups).

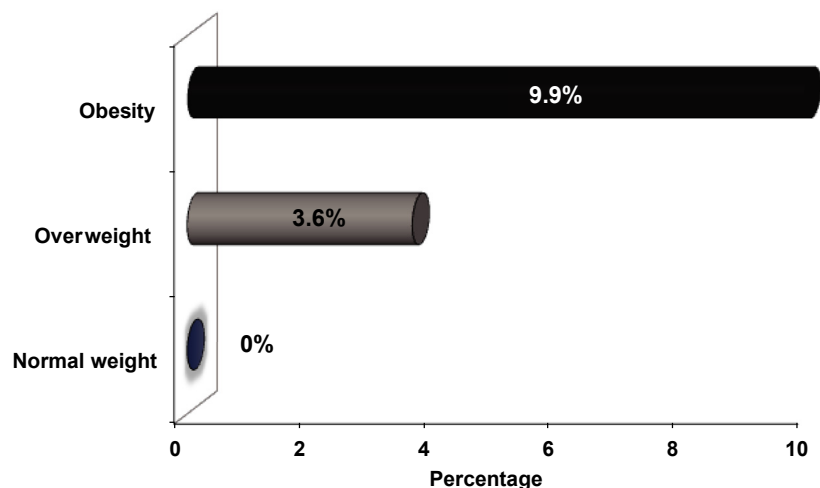


Table 3. Comparison of Clinical and Biochemical Variables According to the Presence of Kidney Alterations

Variable	Kidney Alterations	
	No (N = 163)	Yes (N = 9)
Age (years)	11 ± 3	9 ± 2
Female sex, n (%)	70 (43)	3 (33)
Birth weight (g)	3306 ± 540	3000 ± 846
Systolic blood pressure (mmHg)	105 ± 11	107 ± 7
Diastolic blood pressure (mmHg)	67 ± 9	67 ± 7
Glucose (mg/dL)	86 ± 10	81 ± 10
Uric acid (mg/dL)	4.7 ± 1.3	4.9 ± 1.5
Total cholesterol (mg/dL)	162 ± 31	148 ± 20
LDL cholesterol (mg/dL)	91 ± 27	92 ± 16
HDL cholesterol (mg/dL)	43 ± 11	32 ± 8*
Triglycerides (mg/dL)	134 ± 76	119 ± 71

Conversion factors for units: glucose in mg/dL to mmol/L, $\times 0.05551$; uric acid in mg/dL to $\mu\text{mol/L}$, $\times 59.48$; cholesterol in mg/dL to mmol/L, $\times 0.02586$; triglycerides in mg/dL to mmol/L, $\times 0.01129$. LDL, low-density lipoprotein; HDL, high-density lipoprotein.

* $P < .05$ versus subjects without kidney alterations.

having overweight in early life or developing overweight from puberty to age 20 years was associated to an increased risk of kidney disease in later life.¹⁸

Noteworthy, only subjects with overweight and obesity had early markers of kidney abnormalities in this study. MA was found in 3.2% of children and adolescents with overweight and obesity, which is lower than 10%–15% reported in other studies.^{19–21} The latter studies, however, did not include a control group and were performed in specialized obesity centers, which might represent significant reference bias; in contrast, our study included a group with normal weight, and participants were selected from children and adolescents attending a primary health-care unit. National Health and Nutrition Examination Survey III, one of the largest population-based studies, found a 9.5% prevalence of MA in healthy children and adolescents aged 6–19 years (11% with

obesity); nevertheless, by the nature of that study, it cannot be definitely ruled out that people with a previous kidney disease or transitory MA causes were included.²² In contrast, in a screening national study, 2.1% of cases with MA were found in Japanese school children²³ and 3% in Korean school children²⁴; in the latter study, MA was more frequent in children with obesity than in those with normal weight. Moreover, in our study, GFR alterations were present in 6 cases, from which, 5 displayed hyperfiltration. Hyperfiltration is not strictly considered in the chronic kidney disease (CKD) definition¹³; however, we decided to include it in the analysis as it has been clearly shown as an early marker of kidney disease in conditions such as diabetes, hypertension, and obesity.^{5,25}

Hyperfiltration has also been associated to kidney adiposity and microvascular proliferation in the presence of metabolic syndrome.²⁶ Xiao et al.¹⁹ and Franchini et al.²⁷ found 7% and 16%, respectively, of hyperfiltration in children and adolescents with obesity. It has been shown that children and adolescents with a low or high GFR have higher prevalence of hypertension, left ventricular hypertrophy, and MA than those with the reference category (20–80 percentile of GFR).²⁸ In our study, only 3 patients with hyperfiltration (all of them with obesity) had hypertension but not MA. Results of the Chronic Renal Disease in Turkey study²⁹ showed a general prevalence of 0.13% of decreased GFR in children and adolescents aged 5–18 years, which was more frequent in those with overweight and obesity. In our study, only 1 adolescent with obesity had decreased GFR, which represents 0.58% of the total sample. Contrasting with almost all the previous studies, our study provides information about kidney alterations, considering both kidney damage (MA) and function (GFR), as it is currently recommended.¹³

In addition, subjects with kidney alterations in the present study had higher BMI, body fat markers, lower HDL cholesterol, and a nonsignificant trend to higher systolic BP than those without abnormalities. In adults, all the latter factors are strongly associated with the development of kidney disease.^{2–4} Fat deposition in the glomerulus may alter renal production of vasoactive and inflammatory mediators related to glomerular damages.³⁰ In a cohort study, Hayashi et al.³ showed that subjects with the lowest HDL cholesterol concentration had higher GFR decline over 1 year of follow-up. In addition, subjects with obesity had higher risk for CKD,² and those with central fat distribution have even higher risk than those with peripheral fat distribution.⁶

Sex has been traditionally considered a factor with an influence in kidney function and progression of CKD. Men had larger and wider kidneys, whereas women have higher renovascular resistance, lower absolute GFR, and lower renal plasma flow than men. Women tend to progress to end-stage renal disease at a lower rate than men, regardless of the etiology. In the present study, no differences in

Table 4. Comparison of Anthropometric and Dietetic Variables According to the Presence of Kidney Alterations

Variable	Kidney Alterations	
	No (N = 163)	Yes (N = 9)
Body mass index (kg/m ²)	23 ± 5	27 ± 4
Waist circumference (cm)	82 ± 15	87 ± 10
High-risk waist, n (%)	80 (51)	8 (89)*
Subscapular skinfold (mm)	23 ± 11	35 ± 7*
Body fat (%)	30 ± 10	39 ± 6*
Energy intake (Kcal/day)	1927 ± 621	1833 ± 420
Protein intake (g/kg/day)	1.45 ± 0.73	1.44 ± 0.51
Lipid intake (%)	34 ± 9	32 ± 12
Saturated fat (%)	11.2 ± 4.2	9.7 ± 3.9
Monounsaturated fat (%)	9.9 ± 3.8	8.5 ± 4.6
Polyunsaturated fat (%)	5.5 ± 4.1	5.3 ± 3.3
Cholesterol intake (mg)	250 ± 191	198 ± 146

* $P < .05$ versus subjects without kidney alterations.

gender were found; however, differences in follow-up should be investigated in further studies.³¹

Results of a multivariate analysis support our findings, as higher BMI and lower HDL cholesterol levels were the only significant predicting factors for the presence of kidney alterations. Further investigations with an epidemiological design are needed to confirm our findings and, particularly longitudinal studies, to identify if those children and adolescents with kidney alterations progress to CKD or end-stage renal disease. Notwithstanding, it is noteworthy that in children and adolescents, the presence of overweight and, particularly, obesity was associated with kidney alterations. These findings must alert public health authorities to implement appropriate screening programs for timely detection and treatment of kidney disease and to design prevention programs to avoid obesity and its complications, especially in pediatric population. These actions may help to counteract the current epidemic of CKD in the world.

In conclusion, kidney alterations were observed in 5.3% of the whole sample of children and adolescents of this study. Such alterations were present only in subjects with overweight (3.6%) and obesity (9.9%), who additionally displayed several cardiometabolic and kidney disease risk factors more frequently than those with normal weight. Screening for kidney alterations in high-risk children and adolescents with overweight/obesity may be an excellent opportunity in helping to prevent the burden of kidney failure in adulthood.

Practical Applications

- Obesity was associated to a high proportion of carbohydrate and lipid metabolism alterations, persistence of which may lead to development and progression of chronic diseases, such as CKD, diabetes, and hypertension, in future.
- Screening programs for detection of kidney diseases should start in pediatric population, especially in those subjects with overweight and obesity, not necessarily with the presence of associated hypertension or other metabolic abnormalities. The latter could help to implement adequate management strategies for preventing kidney disease progression and reduce the socioeconomic burden of the disease.
- Strategies for improving nutrition and healthy lifestyles in young population may decrease risk factors for CKD in adulthood.

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